Reverse Brain Drain? Applying Synthetic Control to the DOJ China Initiative's Effect on Chinese Graduate Enrollment in the U.S.

Ella Casano, Michelle Jin, Elsa Zhou

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Guido Imbens and Mary Wootters

1 Introduction

The United States has long derived its power as a global research and innovation hub through its ability to attract foreign talent. Not only have ample studies shown that immigrants represent nearly a quarter of the U.S. workforce in science, technology, engineering, and mathematics (STEM), but immigrants have also disproportionately contributed to U.S. innovation. In particular, China has supplied the most U.S.-based foreign scientists for over two decades (Xie et al. 2023). However, as U.S.-China economic competition continues to amplify, policymakers have expressed increased concerns about economic espionage within the U.S. research apparatus. In response, the Department of Justice launched the China Initiative in 2018, seeking to widely prosecute Chinese-born scientists and university researchers suspected of economic espionage and intellectual property theft.

Not only did departures of Chinese researchers increase by 75% after the China Initiative, but a majority of Chinese college students studying in the U.S. have claimed that this policy stoked an atmosphere of fear and uncertainty for them (Xie et al. 2023). Given that about 25% of all international students in the US come from China, we seek to answer the question: did the Department of Justice's China Initiative reduce subsequent Chinese graduate student enrollment to U.S. universities? This question is compelling because it fundamentally relates to the U.S.'s capability to continue fostering international research collaboration and innovation. The U.S. thrives on its capability as a knowledge power that can attract foreign talent, and a potential dearth of incoming research talent from abroad threatens the U.S.'s robust research apparatus. To answer this question, we use synthetic control as a causal inference tool to construct a counterfactual trajectory for Chinese enrollment, thus evaluating whether the China Initiative had a measurable effect on Chinese enrollment in US universities.

2 Background

Amid rising concerns over research security and economic espionage in an era of intensifying U.S.–China techno-economic competition, the Department of Justice launched the China Initiative in 2018 under the Trump administration. The program aimed to identify and prosecute instances of intellectual-property theft, economic espionage, and undisclosed ties to Chinese institutions, primarily affecting researchers of Chinese descent working in STEM fields. While framed as a national-security measure, the initiative rapidly generated an atmosphere of pervasive anxiety for Chinese students and scholars in the United States. A survey conducted by University of Arizona found that following the initiative, "51% of scientists of Chinese descent ... feel considerable fear ... about being surveilled" by the U.S. government (Guo et al. 2021).

Beyond concerns about the China Initiative stoking disproportionate fear among Chinese and Chinese-American communities in the United States, the DOJ initiative also received much scrutiny from policymakers and civil rights advocates over its lack of strategic direction and damage to international research collaboration. While the initiative began as a directive to prosecute economic espionage, it soon began to prosecute researchers with broadly-defined research integrity concerns, such as failing to fully disclose foreign affiliation on forms. Only a third of cases across 150 defendants

in 77 cases actually resulted in conviction (Guo et al. 2021). Although the China Initiative ended in February of 2022 under President Joe Biden, the flow of Chinese talent to the U.S. has continued to be threatened by subsequent policies, such as a series of investigations against Chinese researchers on research integrity concerns under the False Claims Act in 2025 (Gorelick et al. 2025).

3 Data

For our analysis, we used public data from Open Doors, an information resource produced by the Institute of International Education and sponsored by the U.S. Department of State that collects enrollment of international students reported by roughly 1,600 institutions. We filtered the data to capture annual international graduate student enrollment numbers aggregated by country of origin, from 2014 to 2024.

In addition to enrollment, we included covariates using data published by the World Bank, again aggregated by country, on GDP per capita, share of the male population aged 20-24, and share of the female population aged 20-24.

4 Methods

This study employs a synthetic control approach (Abadie et al. 2010) to simulate a counterfactual trend for China's graduate enrollment in U.S. institutions after the China Initiative took effect (2019 and onward). Our research question is broadly consistent with the requirements for synthetic control. First, the data consist of aggregate data at the country level with a single treated unit, China. We satisfy convexity by first calculating graduate enrollment growth rates year-over-year, and then applying a log transformation. In addition, consistent with Abadie (2021), before implementing our synthetic control analysis, we selected a smaller donor pool from the full set of countries in the Open Doors dataset. We constructed this donor pool through similarity in observed attributes and suspected similarity in unobserved attributes, particularly cultural attitudes toward studying in the U.S. and COVID-19 policy approaches. Our donor pool for our synthetic control included South Korea, Vietnam, India, Japan, Singapore, and Thailand.

We may expect some spillover effects to additionally impact our results if we expect that U.S. institutions find Chinese graduate students substitutable with graduate students from countries in this donor pool. However, we suspect that substitution is imperfect, for a variety of reasons. One is student quality, which may be significantly higher for countries with larger populations due to intensified competition for admission like China, reducing substitution for the smaller and less competitive countries in our donor pool. In addition, admissions offices (especially since we are evaluating graduate enrollment, rather than undergraduate) may be risk-averse to a certain extent with regard to accepting students from more unfamiliar international institutions, as there is reduced signaling of student quality, academic rigor, and research acumen. This may impact the ability to substitute away from well-established Chinese institutional pathways to new ones.

4.1 Synthetic control model

We have data for $J+1$ units, where $j=1$ is the treated country, China, and $j = 2, ..., J+1$ is the donor pool of untreated countries. Our data spans $T = 10$ years, where T_0 represents the pre-period of 2014-2018. For each country j and year t, we observe Y_{jt} , the log enrollment growth rate. We also observe k predictors for each country $j, X_{1j}, ..., X_{kj}$, which include the log enrollment growth rate for each year in T_0 and our covariates log GDP per capita, share of men aged 20-24, and share of women aged 20-24.

We define Y_{jt}^N to represent the log enrollment growth rate for each country j and year t without intervention, and Y_{jt}^I to represent the log enrollment growth rate with intervention. For our treated country $j = 1$, where we do observe Y_{1t}^I, we need to estimate Y_{1t}^N for our post-period, or $t > T_0$. This will allow us to express the effect of the intervention as the following:

$$\tau_{1t} = Y_{1t}^I - Y_{1t}^N \tag{1}$$

Formally, the synthetic counterfactual is a weighted average of the donor pool countries represented by a $J \times 1$ vector of weights, $\mathbf{W} = (w_2, ..., w_{J+1})'$. We can express the following:

$$\hat{Y}_{1t}^N = \sum_{j=2}^{J+1} w_j Y_{jt} \tag{2}$$

$$\hat{\tau}_{1t} = Y_{1t} - \hat{Y}_{1t}^N \tag{3}$$

We choose $\mathbf{W*} = (w_2^*, ..., w_{J+1}^*)'$ that minimizes

$$\|X_1 - X_0 W\| = \left(\sum_{h=1}^{k} v_h (X_{h1} - w_2 X_{h2} - \cdots - w_{J+1} X_{hJ+1})^2 \right)^{1/2} \tag{4}$$

while continuing to follow Abadie (2021) by requiring the weights $w_2, ..., w_{J+1}$ to be non-negative and sum to one.

The vector $\mathbf{V} = (v_1, ..., v_k)$ represents weights that determine the relative importance of matching each predictor between China and the synthetic counterfactual. These weights represent how strongly the optimization prioritizes fitting each predictor in constructing the synthetic control.

Since we rely on the **synth** package in Stata, we choose $\mathbf{V*}$ by minimizing mean squared prediction error (MSPE) for the pre-period as follows:

$$\sum_{t \in T_0} \left(Y_{1t} - \sum_{j=2}^{J+1} w_j(V) Y_{jt} \right)^2 \tag{5}$$

In sum, we find the synthetic control $\mathbf{W*} = W(V^*)$ by minimizing the MSPE. We express the treatment effect as τ_{1t}, averaged in our post-period to calculate the ATT.

5 Results

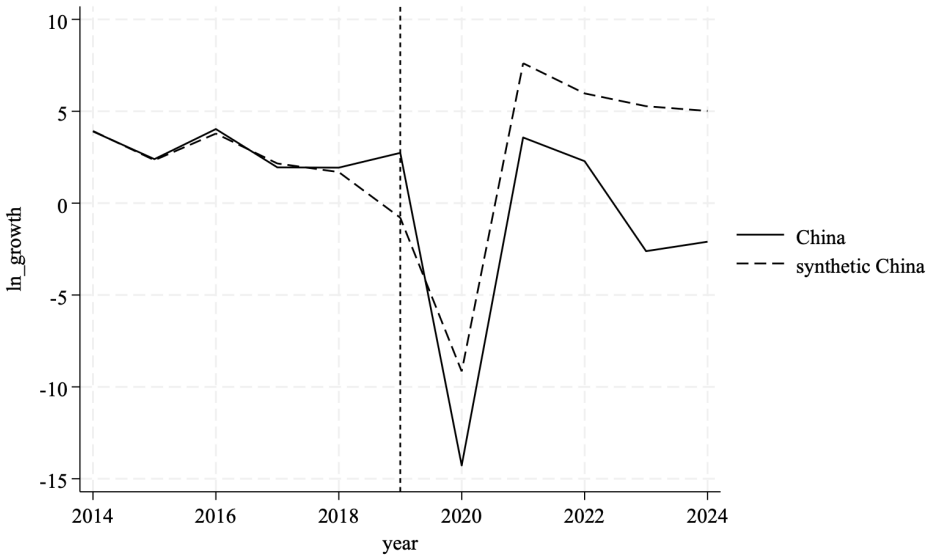

Fig. 1 Trends in log enrollment growth: China vs. synthetic China.

Figure 1 displays log enrollment growth rate for China and its synthetic counterfactual from our evaluation period, 2014-2024. Synthetic China and actual observed values of China are very similar in the pre-period but diverge noticeably 2019 and onward. We can look at the predictor means of China versus its synthetic in Table 1, which reveal that they are mostly matched. This gives us confidence that our synthetic China is a sensible approximation of the log growth rate of Chinese graduate enrollment in U.S. institutions post-2019, had the China Initiative not occurred.

Table 1 Enrollment predictor means

Predictor	**China**	**Synthetic China**
ln(gdp)	9.0617	8.9022
Share of male 20-24	7.0839	8.1691
Share of female	6.5538	7.5623
$\ln(\text{growth})_{2014}$	3.9013	3.9266
$\ln(\text{growth})_{2015}$	2.3969	2.3379
$\ln(\text{growth})_{2016}$	4.0311	3.7867
$\ln(\text{growth})_{2017}$	1.9471	2.1587
$\ln(\text{growth})_{2018}$	1.9324	1.6899

Table 2 displays the weights on each donor pool country for the synthetic counterfactual. These results indicate that the log growth rate of Chinese graduate enrollment pre-China Initiative is best approximated with a linear combination of trends from South Korea, Vietnam, Sri Lanka, India, and Singapore.

Table 2 Weights

Country	**Weight**
Japan	0.000
South Korea	0.297
Vietnam	0.301
Sri Lanka	0.342
Thailand	0.000
India	0.015
Singapore	0.044

The treatment effect we set out to measure is the difference between the observed trends for Chinese graduate enrollment and the synthetic counterfactual after the China Initiative was launched. While both trendlines experience a dip at the outset of the pandemic and a quick recovery, the synthetic trendline recovers noticeably more sharply, suggesting that the observed values for Chinese graduate enrollment might reflect a negative impact caused by the China Initiative.

We find that the average treatment effect over the post-period 2019-2024 is -4.05%. In other words, we estimate that the China Initiative caused a 4.05% drop in Chinese graduate enrollment after its launch. The magnitude of this gap seems to increase over time. Figure 2 shows the gap between the observed values for log enrollment growth rate for Chinese graduates and the synthetic version. Reasonably, the pre-period gap is near 0, while the post-period gap grows larger over time.

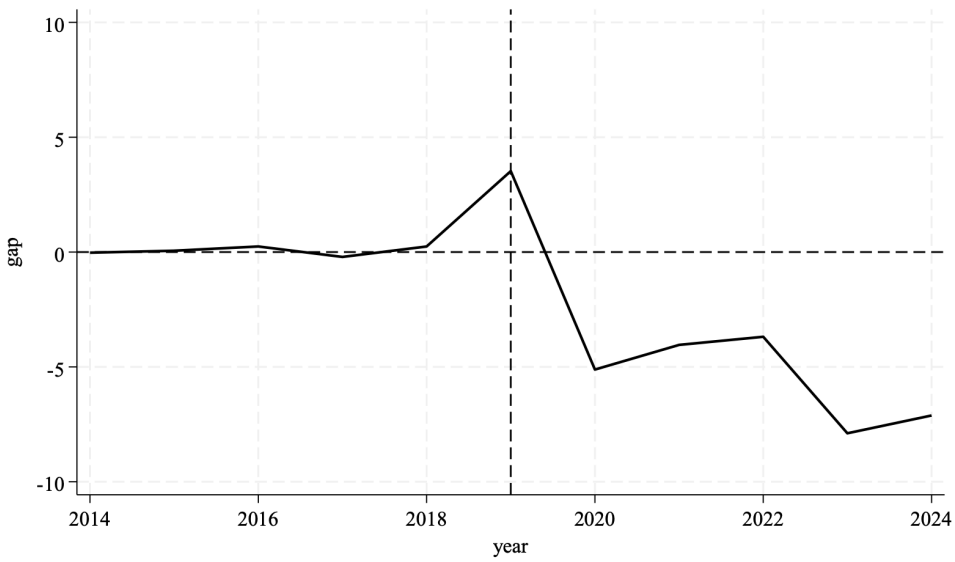

Fig. 2 Log enrollment growth rate gap between China and synthetic China.

6 Robustness checks

Table 3 Placebo analysis: MSPE

Synthetic country	RMSPE	MSPE ratio
China	0.1819	902.3826
Japan	1.7910	34.2096
South Korea	2.2692	13.9099
Vietnam	2.5621	13.6971
Sri Lanka	2.5574	6.9329
Thailand	5.6867	4.4126
India	12.1433	4.2147
Singapore	4.3606	1.2063

To assess whether our estimated treatment effect is actually significant, we conduct a standard placebo test as defined by Abadie et al. (2010). This test involves creating synthetic counterfactuals where each country in our donor pool is handled as the treated unit, and then computing the MSPE ratio as a measure of fit in the pre- versus post-period. If the gap between China and its synthetic is large, we expect a high MSPE ratio. Further, if the gap between China and its synthetic is large, and the gap between other countries' and their synthetic versionsn are much smaller, we have confidence that China faced an additional factor impacting enrollment that the

other countries in the donor pool did not. Since we selected our donor pool based on similarities in observed and suspected unobserved attributes, we would have increased confidence that the China Initiative was the additional factor.

Indeed, in the placebo distribution shown in Table 3, China's post-treatment MSPE ratio far exceeds those of all donor countries, the second of which is Japan. This large gap indicates that China's post-2019 trajectory is uniquely inconsistent with its pre-treatment synthetic match and is suggestive that China faced an additional factor, the China Initiative, which impacted enrollment in China but not the other countries in our donor pool.

We also conduct leave-one-out analysis as defined by Abadie et al. (2015) as a secondary form of robustness analysis to ensure that the estimated effect is not driven by a single outlier. The leave-one-out analysis operates by re-computing the post-period ATT using the original donor pool minus one country. Encouragingly, we find that removing any individual donor country produces similar post-period ATTs mostly clustered tightly around -4.05%. Only the exclusion of Vietnam shifts the estimate meaningfully to about -5.31%. The importance of Vietnam in the synthetic version of China may be due to the highly overlapping zero-COVID policies pursued by both governments at the outset of the pandemic (Barnes 2022).

Table 4 Leave-one-out analysis

Country left out	**ATT post**
China	-4.0535
Japan	-4.0535
South Korea	-3.8735
Vietnam	-5.3143
Sri Lanka	-3.1438
Thailand	-4.0535
India	-3.8476
Singapore	-4.1928

Our analysis is constrained by a few limitations: the confounding effects of the pandemic, the use of total rather than first-year enrollment data, the inability to capture sector-specific impacts, and the reliance on a reported survey. Since the pandemic began shortly after our treatment year, it greatly impacted graduate enrollment and immigrant outflow between all countries. While we attempt to blanket the impact of the pandemic by selecting donor countries that mirror the COVID-19 policies of China (our unit of interest), the pandemic did not impact all countries equally. Our data also was limited to total enrollment, which means that we cannot tell the difference between new entrants and continuing students, which weakens causal inference because the China Initiative likely affected new entrants the most. Our data also did not include graduate enrollment by academic subject, meaning that we could not isolate the impact of the China Initiative on STEM graduate enrollment, the subject

most impacted by the policy. Finally, the dataset is survey-reported across 3000 institutions, meaning it may miss students in non-traditional programs and institutions that did not participate in the survey.

7 Conclusion

Our results find that the China Initiative dropped Chinese graduate enrollment in U.S. universities by approximately 4.05%. While the impact may be overstated due to the limitations explained above, our results indicate that at least drawing a causal negative link between the China Initiative and Chinese graduate enrollment is robustly supported. In the future, we can attempt to seek additional precise data that addresses our limitations, such as by isolating first-year STEM enrollment for doctoral applicants rather than total-year generic graduate enrollment. We should also chart and incorporate the effect of other coinciding policy shocks after 2019. As U.S.-China relations continue to oscillate, actions such as the escalation of U.S.-China trade war in 2019 and the occurrence of record-high visa rejections in 2023 all impact the flow of Chinese international students from China to the United States — a critical concern for the U.S. research apparatus that we ought to continue researching.

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